ELEMENT COMBINATION RULE

CIRCUIT ANALYSIS METHOD

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Atomic Structure

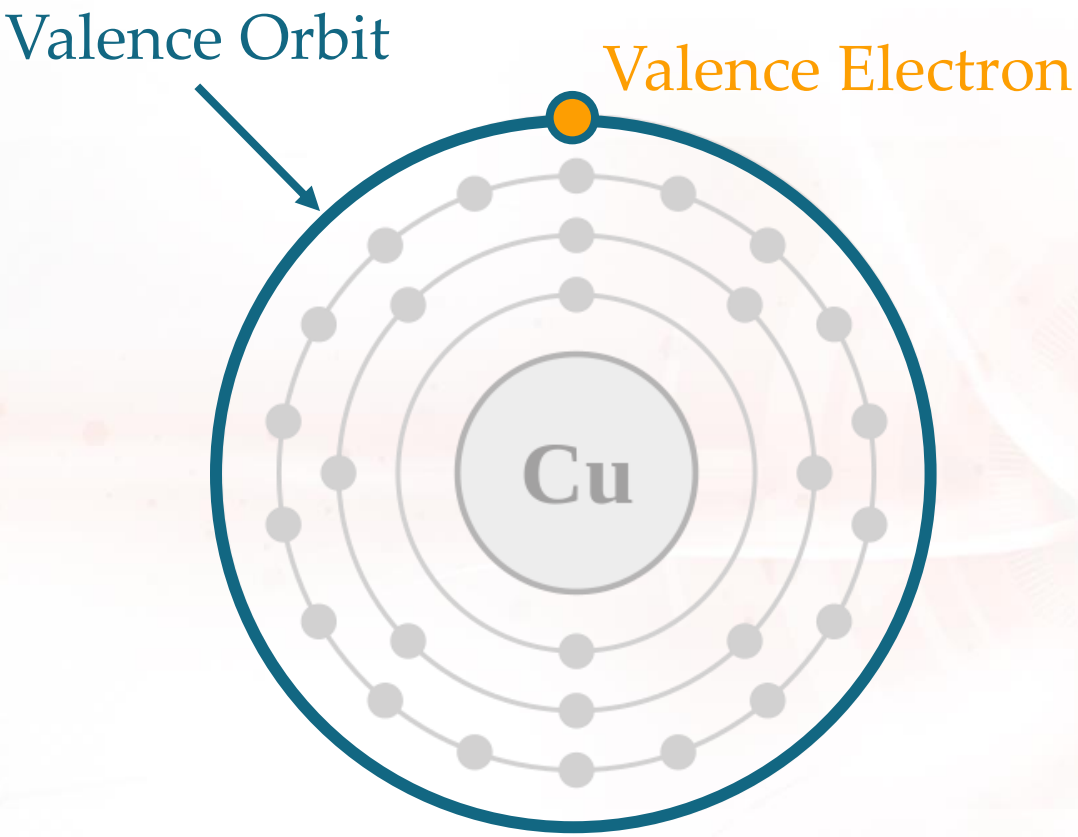
Ohm's Law

Series Network

Parallel Network

ATOMIC STRUCTURE

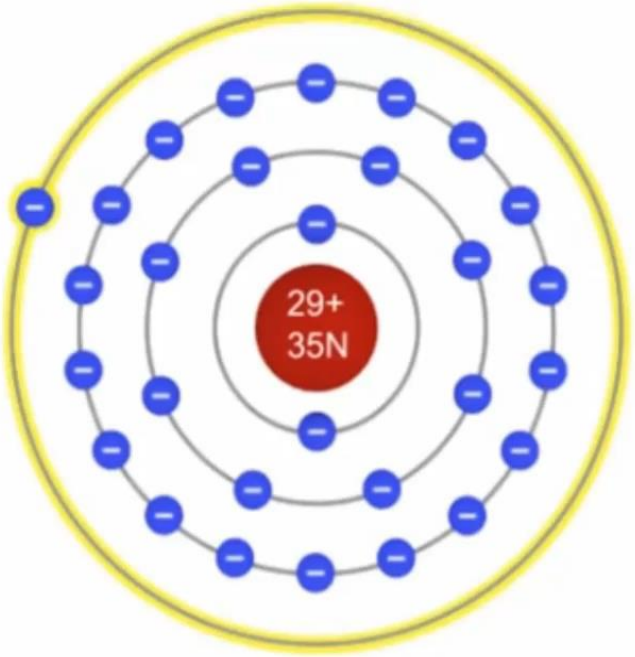
VALENCE ORBIT



The **outermost** electron shell or energy level of an atom

Controls the **electrical properties** of the atom

CONDUCTOR



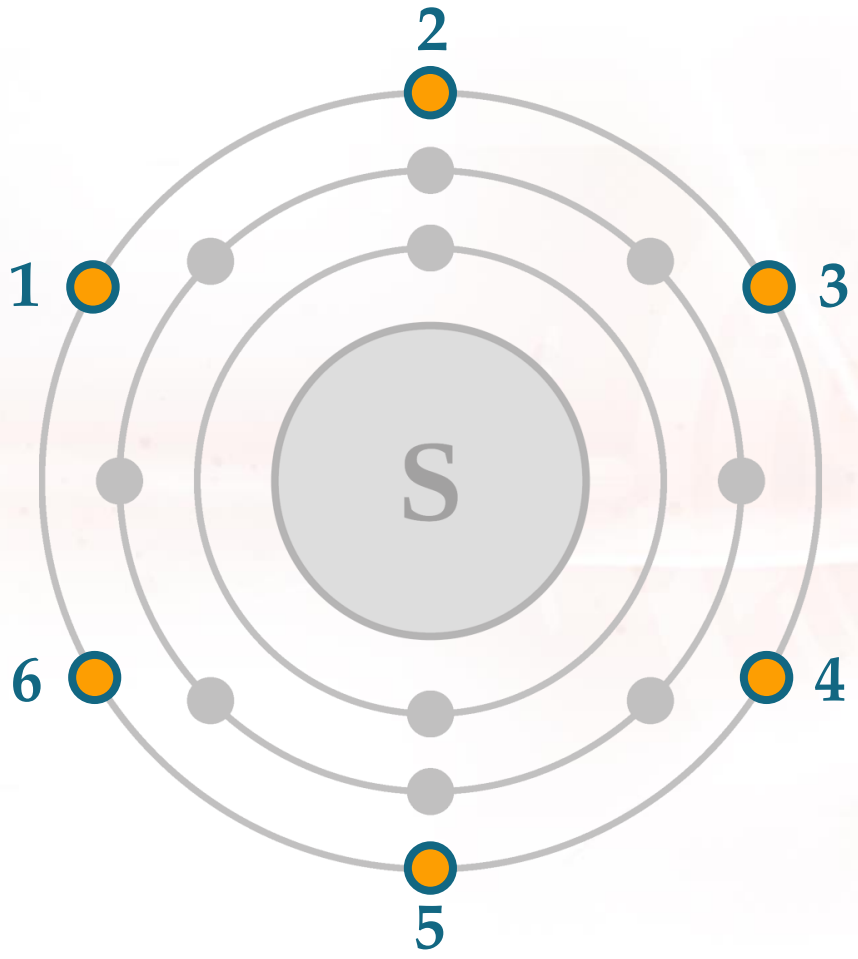
Materials that **allow** the electric current to flow through them

- **Copper**
- Silver
- Gold

Valence Electrons

- **Less** than **4** electrons

INSULATOR



Materials that **do not** conduct electricity

Plastic

Rubber

Sulfur

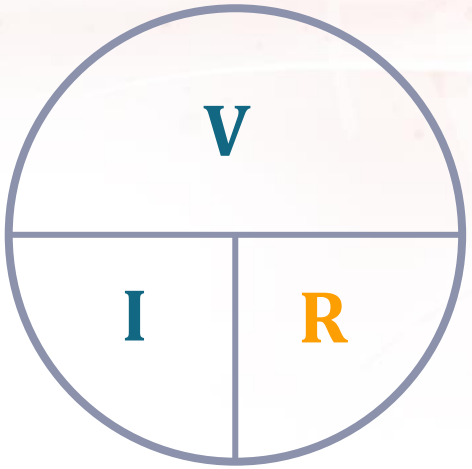
Valence Electrons

More than **4** electrons

OHM'S LAW

The ratio of voltage to current is

constant



$$R = \frac{V}{I}$$ unit: Ohm (Ω)

$$V = IR$$ unit: Volt(V)

$$I = \frac{V}{R}$$ unit: Amp (A)

EXERCISE

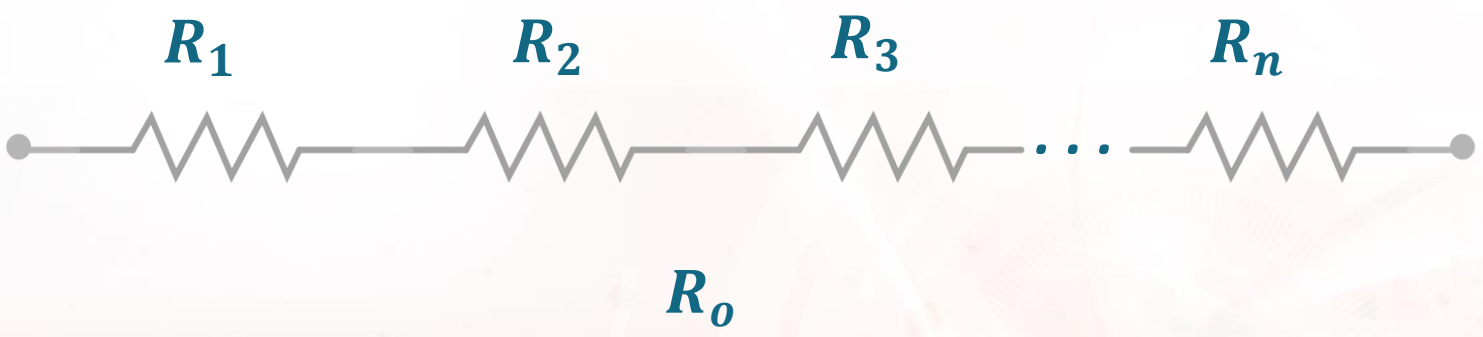
What would be the **current** flowing through a circuit with **4 ohms** resistance when a voltage of **12 volts** is applied?

EXERCISE

What is the **voltage** drop across a circuit with a resistance of **5 ohms** when a current of **2 amps** flows through it?

SERIES NETWORK

TOTAL RESISTANCE

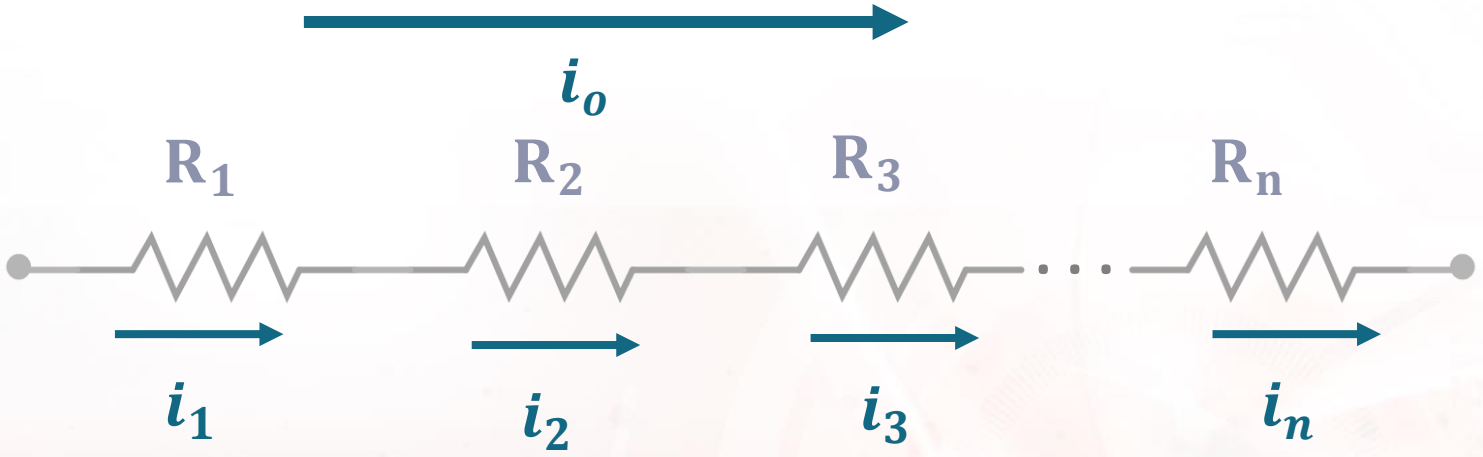


The Total Resistance, R_o

Resistance **adds** in series

$$R_o = R_1 + R_2 + R_3 + \ldots R_n$$

TOTAL CURRENT

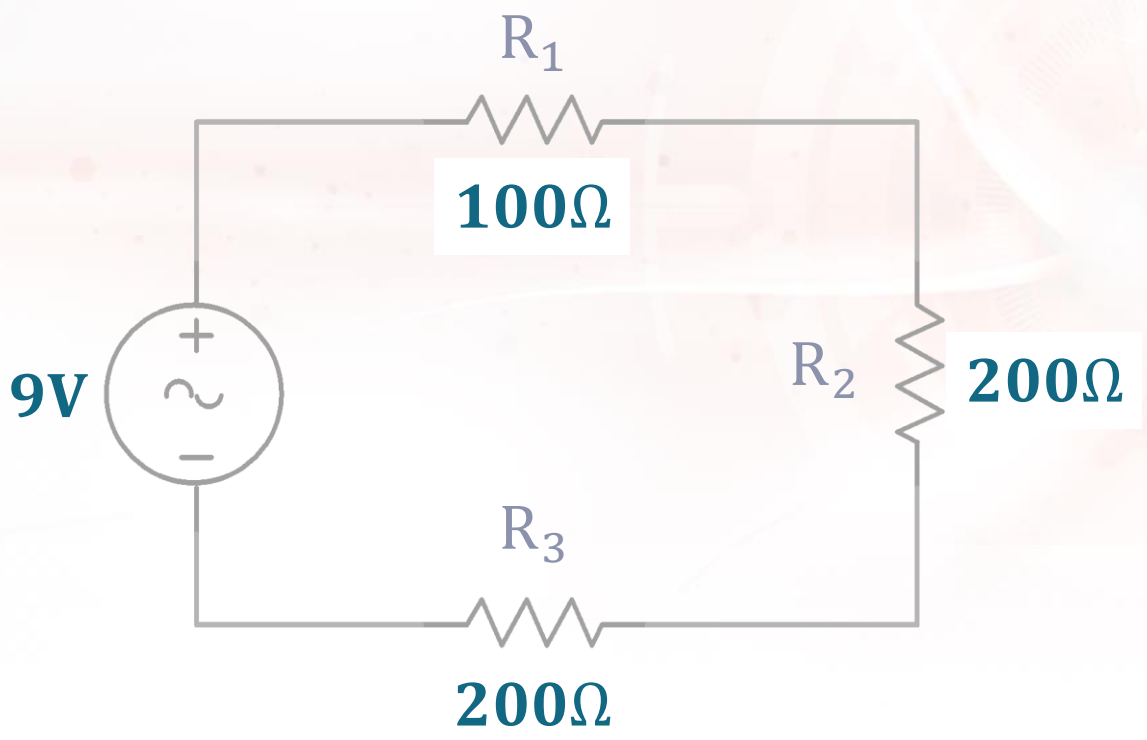


The Total Current, i_o

Currents are **equal** in series

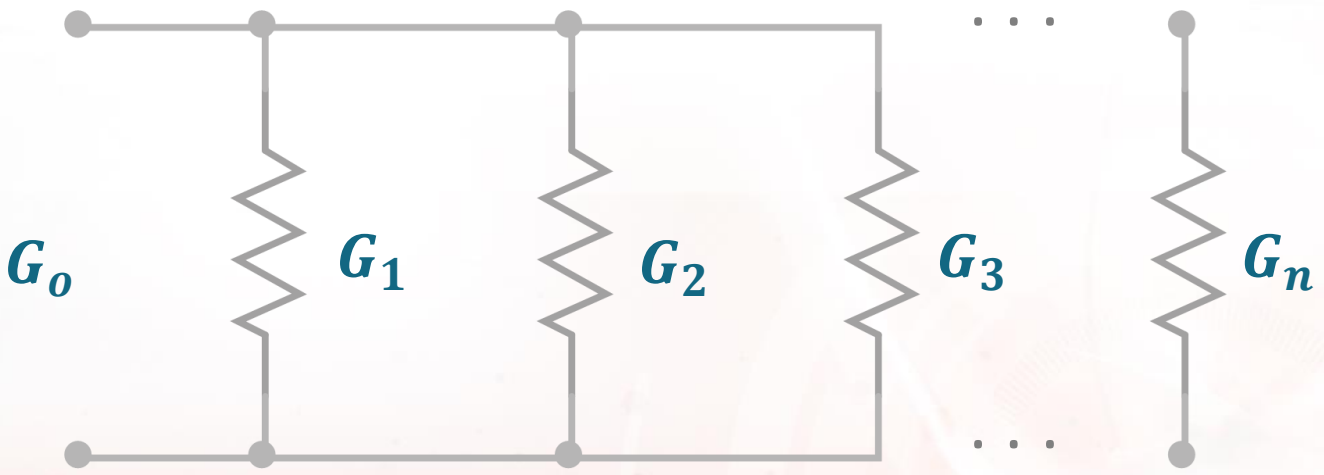
$$i_o = i_1 = i_2 = i_3 = \ldots i_n$$

Determine the **total resistance**, the **total current**, and the **voltage** drop of each resistor in the given **circuit**.



PARALLEL_NETWORK_

TOTAL CONDUCTANCE



note:

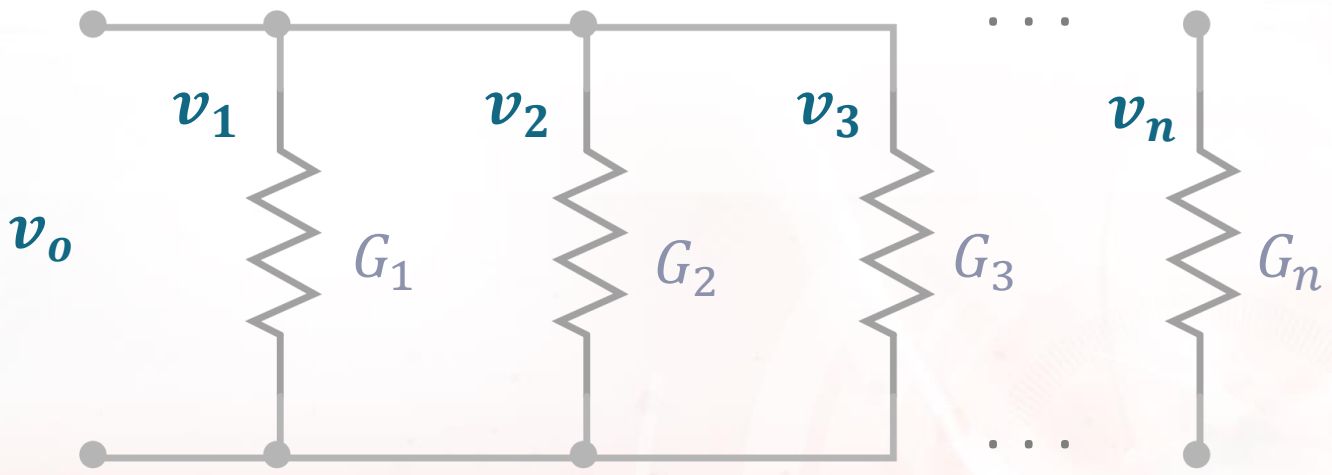
$$G = \frac{1}{R}$$

The Total Conductance, G_o

Conductance **adds** in parallel

$$G_o = G_1 + G_2 + G_3 + \ldots G_n$$

TOTAL VOLTAGE

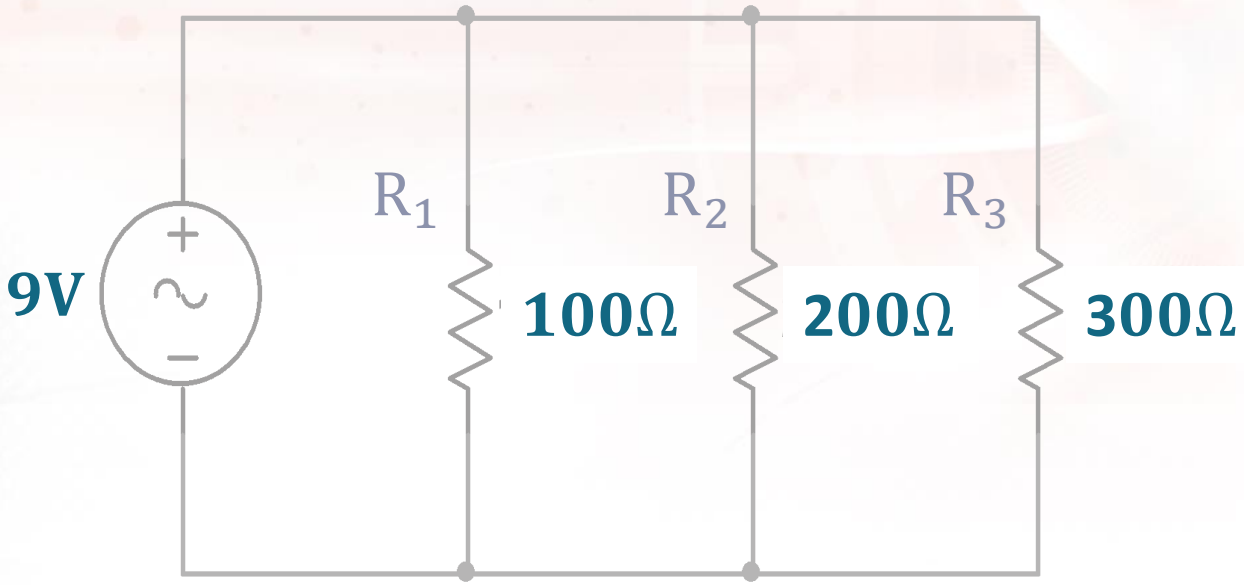


The Total Voltage, v_o

Voltages are **equal** in parallel

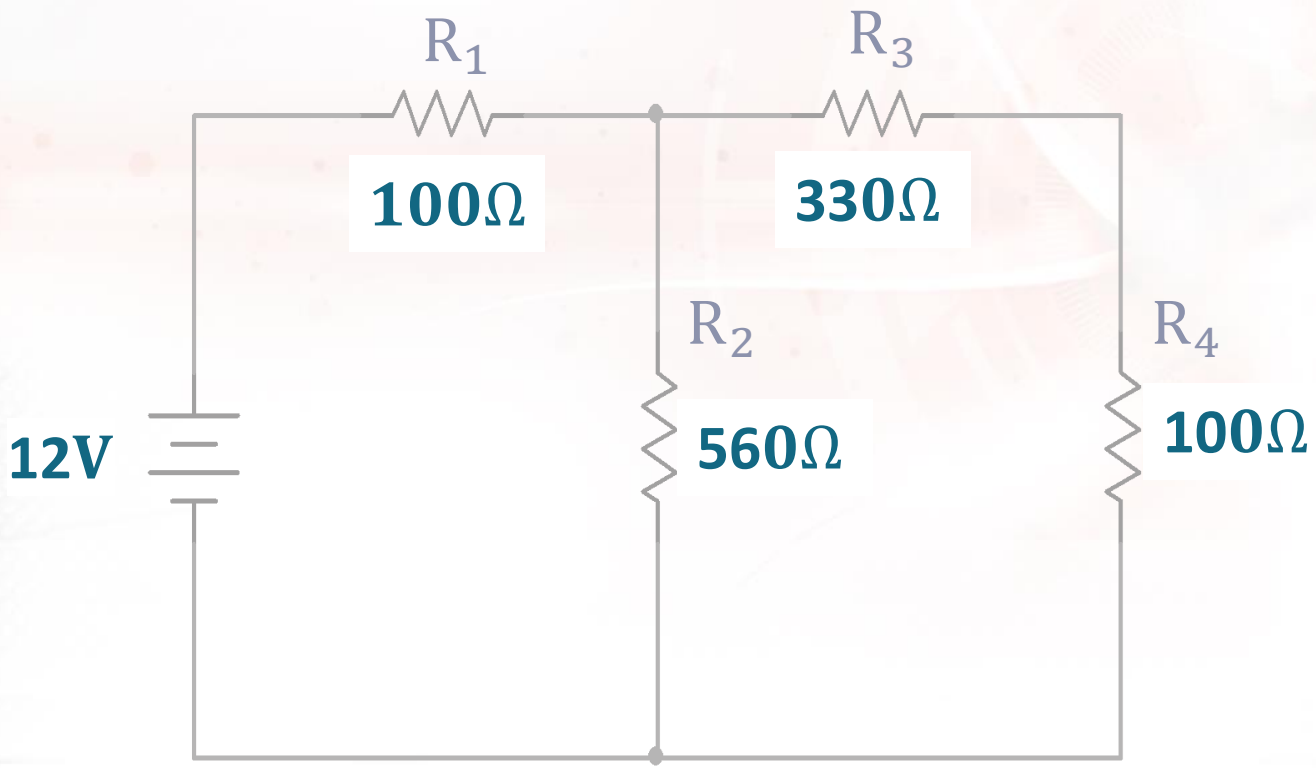
$$v_o = v_1 = v_2 = v_3 = \ldots v_n$$

Determine the **total resistance**, the **total current**, and the **current** flowing through each resistor in the given **circuit**.



Series	Parallel
Resistance adds in series	**Conductance** adds in parallel
Currents are equal in series	**Voltages** are equal in parallel

Determine the **total resistance**, the **total current**, and the **current** flowing through each resistor in the given **circuit**.



LABORATORY